A Point-Set Topology-Based Information Entropy Estimation Method for Hyperspectral Target Detection

Xiaotong Sun, Lina Zhuang, *Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Hongmin Gao, *Member, IEEE*, Xu Sun, *Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*— With hyperspectral remote sensors (imaging spectrometers) imaging a scene, the specificity of the target of interest is manifested in the significant differences between it and the surrounding background in terms of quantity, spatial distribution, and spectral characteristics, which provides conditions for the implementation of pixel-level diagnostics for target detection. Traditional model-driven methods utilize specific model assumptions to parse hyperspectral image (HSI) data in scenes with variability and are prone to encounter limitations due to model-data discrepancy. Most data-driven methods are limited in practical applications due to the great demand for training samples, the large number of parameters to be determined, and the costly computational complexity. To address the limitations of the existing methods, this article adopts point-set topology theories to analyze the properties of hyperspectral data at the mathematical-statistical level and seek a solution for the information retrieval task of target detection, whereby a target detection method through information entropy estimation based on point-set topology is proposed. First, parallel topological spaces are constructed to order the original HSI data to ensure that the differences in data features between various classes of land covers are reflected in intuitive properties in the topological spaces. Second, in conjunction with the priori information about the target, information entropy estimation is introduced to select optimal separable spaces for the target and the background by measuring the degree of ordering of data to achieve an accurate separation. Finally, a proper way to quantify and highlight the differences in data features between various land covers in the optimal separable spaces is explored for the algorithmic output to perform the information retrieval task. The proposed target detection through information entropy estimation based on point-set topology (TD-IEEPST) exploits an innovative combination of point set topology theories and information entropy estimation to achieve efficient extraction of land cover information for detection, ensuring both theoretical interpretability and computational efficiency. Extensive experimental results on real hyperspectral datasets verify that the proposed method is ahead of other widely used and state-of-the-art methods in terms of computational cost, detection effects, and robustness, and promising to provide technical support for detection response requirements in practical applications.**

***Index Terms*— Hyperspectral image (HSI), information entropy, machine learning, target detection, topology.**

I. Introduction

COMPARED with traditional imaging approaches, the most prominent advantage of hyperspectral remote sensing is that it integrates both spatial and spectral information for land covers within a specific scene [1], [2], [3]. Due to the different physical and chemical properties of various classes of land covers, there are subtle differences in spectral characteristics [4], [5], based on which hyperspectral remote sensing is capable of accurately identifying objects that cannot be recognized by ordinary optical remote sensing techniques [6], [7]. Hyperspectral target detection is a technology with strong comprehensiveness and high research value, and the related applications cover a wide range of fields for both civilian and military purposes [8], [9]. It has already demonstrated unique advantages in geological surveys, precision agriculture and forestry, damage assessment, camouflage exposure, and other aspects [1], [10], [11]. Subsequently, hyperspectral target detection will continuously provide critical technical support for various geoscientific and national defense and security applications [12], [13].

Since the 1990s, scholars have adopted theories and models in the field of signal processing to achieve the processing and analysis of hyperspectral remote sensing images [14], [15]. In terms of the search for solutions for target detection, there are well-developed design frameworks for detectors as well as algorithmic systems [16], [17]. Among them, the detection methods developed based on the probability statistics (whited-space) model and the subspace projection model have been most widely used [14], [18]. The design framework of the former can be abstracted as binary hypothesis testing in statistics [6]. With the mathematical modeling of target

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Xiaotong Sun is with the College of Information Science and Engineering, Hohai University, Nanjing 211100, China, and also with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: sxt1996@outlook.com).

Lina Zhuang, Lianru Gao, and Xu Sun are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhuangln@aircas.ac.cn; gaolr@aircas.ac.cn; sunxu@aircas.ac.cn).

Hongmin Gao is with the College of Information Science and Engineering, Hohai University, Nanjing 211100, China (e-mail: gaohongmin@hhu.edu.cn).

Bing Zhang is with the College of Information Science and Engineering, Hohai University, Nanjing 211100, China, also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zb@radi.ac.cn).

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detection problems, it is usually assumed that the testing pixels obey their respective multidimensional normal distributions when the two assumptions are valid [19]. The critical technical point here is to sufficiently mine the information of the target and the background in real scenes and accurately estimate the model parameters to meet the requirements of high-precision detection. The representative algorithms are matched filter (MF) [20], constrained energy minimization (CEM) [15], generalized likelihood ratio test (GLRT) [21], adaptive coherence/cosine estimator (ACE) [22], adaptive MF (AMF) [23], etc. In addition, the subspace projection transformation is an important mathematical tool, and the related models are widely used in the design of detectors [24]. The limitation of spatial resolution and the complexity of the distribution of land covers in imaging scenes lead to the fact that in most cases, the spatial coverage of each pixel may include different substances, including the target of interest and the background, from which a mixed model containing spectral signals of different substances can be derived. Therefore, target detection using subspace projection transformation aims at constructing various forms of target or background subspaces to accurately represent the testing pixels in hyperspectral images (HSIs), to highlight the target, suppress the background, and maximize the signal-to-noise ratio (SNR) to achieve ideal detection effects. The classic orthogonal subspace projection (OSP) and its various derivative versions such as signature subspace projection (SSP) [24], target subspace projection (TSP) [25], oblique subspace projection (OBSP) [26], etc., have been verified to be robust. In particular, there are algorithms that incorporate the advantages of both models and are widely used, such as target-constrained interference minimized filter (TCIMF) and OBSP-based GLRT (OGLRT) [27], [28]. Traditional methods demonstrate the advantages of less demand on the quality of training samples, low time complexity, and ease of development and integration, which enable them to respond quickly to the detection requirements under the conditions of limited computational resources and training samples. However, with the rapid development of hyperspectral remote sensing imaging techniques, the quality of the acquired data is increasingly high, and the traditional methods are inevitably constrained by specific model assumptions due to their effect mechanism, which makes it difficult to adapt to the variability of real scenes and thus leads to deterioration of detection effects.

Considering the bottlenecks encountered by the traditional model-driven methods in the development, more and more scholars focus on machine learning to seek new breakthroughs and vitality for the field of hyperspectral target detection [29], [30]. Machine learning is a comprehensive interdisciplinary discipline, that has been rapidly developed in the last 30 years [30]. With the rapid development of machine learning theories, the algorithms devised accordingly have been widely applied in various fields including hyperspectral remote sensing [9], [29]. The numerous applications of machine learning in hyperspectral target detection have demonstrated prominent parsing capabilities for data with complexity and achieved promising detection effects [10], [31]. Representation reconstruction, metric learning, kernel mapping, tensor decomposition, and deep learning are several research hotspots that have received wide attention [6]. Specifically, the representation-based methods can be subdivided according to sparse, collaborative, and low-rank representations. Sparse representation-based detector (SRD) [32], combined sparse and collaborative representation-based detector (CSCR) [33], and sparse and low-rank matrix decomposition (SLRMD) represent testing pixels by constructing various forms of target or background dictionaries to achieve a more accurate target-background separation [34]. Besides, a scalable subspace clustering method integrating the learning of a concise dictionary and robust subspace representation in a unified model is proposed, and a new adaptive spatial regularization for the representation coefficients is introduced to incorporate spatial information of HSI and improve the robustness of the model to noise [35]. This provides a potential noise reduction solution for hyperspectral anomaly detection [36]. Methods for metric learning classes can be subdivided into several categories such as support vector machine [37], clustering [38], graph theory [39], and topology [40]. Such methods are free from the constraints of specific model assumptions and enable adaptive learning of the optimal separation between the target and the background to fully exploit and accurately quantify the differences in spectral characteristics of various land covers [41]. Tree-structured encoding-based target detection method [42], ensemble-based information retrieval with mass estimation (EIRME) method [43], and chessboard-shaped topology-based target detection (CTTD) method can realize the optimal target-background separation through hierarchical structures such as binary tree and chessboard [44]. Kernel-based detectors map the original HSI data into a high-dimensional space through a specific form of kernel function to improve the separability between the target and the background [45]. Representative kernel OSP (Kernel-OSP) [46], kernel matched subspace detector (Kernel-MSD) [47], and kernel TCIMF (Kernel-TCIMF) [48], etc. have demonstrated superior processing capabilities for nonlinear data. Tensor decomposition-based methods extract information on land covers for detection by modeling the HSI data cube as a higher-order tensor [49]. Methods such as tensor-matched subspace detector (TMSD), canonical polyadic MF (CPMF), and coarse-to-fine HSI target detector based on low-rank tensor decomposition (HTDLTD) acquire robust performance by taking into account effective information in both spatial and spectral dimensions [50], [51], [52]. In addition, there are methods that employ the fractional Fourier transform (FrFT) as a preprocessor to obtain features in an intermediate domain between the original reflectance spectrum and its Fourier transform with complementary strengths by space-frequency representations [53], [54]. Such methods can be used to remove noise and thus enhance the discrimination between anomalies and background [55]. As an important branch of machine learning, deep learning has been developing rapidly in recent years and excels in a variety of complex and difficult intelligent recognition tasks [56], [57], [58], [59]. The development of deep learning is based on the concept of artificial neural networks, which utilize existing data to construct a massive number of samples to train

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a multilayer network structure to perform classification or recursion tasks on unknown data [60]. Among the variety of network structures developed in deep learning, convolutional neural network (CNN) has achieved promising performance in the field of image processing. CNNs and their extended networks have been increasingly applied to target detection in HSIs [61]. In addition, a series of algorithms derived from autoencoder (AE) [62], [63], deep belief network (DBN), and generative adversarial network (GAN) can extract and exploit the land cover information contained in HSI data [64], [65]. The unique advantages of deep learning in processing diverse, multilevel and high-complexity data have been fully exerted and demonstrated. In summary, hyperspectral target detection based on machine learning has emerged a large number of algorithms with excellent performance, and the theoretical system is gradually maturing. For feature mining in real data with complexity, the machine learning-based methods can fully explore and utilize the abundant detail information provided by HSIs to achieve high-precision target detection. However, the popular machine learning methods mentioned above usually need to meet certain conditions to take effect and achieve satisfactory detection effects. Specifically, the selection of kernel functions, the construction of target or background dictionaries, and the preparation of training samples can directly or indirectly affect the subsequent detection effects. Basically, the validate conditions of these methods are quite strict and often require the consumption of a large amount of time and space resources. Hence, their utilization and popularization in practical applications are restricted to a certain extent, exhibiting limitations.

In summary, the stability and computational efficiency of the traditional methods are undeniable. However, in most cases, machine learning-based methods are ahead of signal processing-based methods in detection effects at the cost of execution time, memory consumption, and other computational resources, which is in line with the current research status and trend of hyperspectral target detection. In this article, the limitations encountered by the existing methods are thoroughly considered. Hyperspectral target detection where priori information about the target is available is treated as an information retrieval task, with a focus on point-set topology theories to seek solutions. The mathematical-statistical properties of hyperspectral data are analyzed to be in direct response to the detection requirements for HSIs. Meanwhile, the information entropy theories are introduced to efficiently extract the information about land covers for detection. As a result, the proposed method balances theoretical interpretability while effectively saving computational resources during execution.

In this article, the HSI being processed is treated as an information collection and the known target spectrum is treated as a given query. In an HSI containing multiple classes of land covers, pixels with high similarity to the given query belong to the target class, and pixels with low similarity to the given query belong to the background class. Thus, hyperspectral target detection can be treated as an information retrieval task in machine learning, which is solved with point-set topology theories. The proposed target detection through information entropy estimation based on point-set topology (TD-IEEPST) first maps the original HSI dataset into parallel topological spaces through a specific form of framework so as to achieve an ordered arrangement of the data, and such an entropy-reduction process is a preparation for unraveling the deep-level data features. Given that, an HSI dataset selected for target detection satisfies the condition of uneven distribution of land covers, i.e., the target and the background belong to the sparse and dense sets, respectively, which are latent and hidden features in the primitive feature space. After mapping, the above data features are reflected in differences in relative locations and densities between the rearranged target and background in the parallel topological spaces, whereby the target-background separation is ensured. In other words, latent and hidden data features in the primitive space are transformed into intuitive properties in the topological spaces, which substantially reduces the difficulty of retrieving the target of interest for the information retrieval task. Based on this, information entropy estimation is introduced to measure the degree of ordering for a system as the topological space, whereby the topological spaces with low uncertainty in the rearrangement of the pixels in the HSI, i.e., topological spaces with a strong separability between the target and the background are selected as the optimal separable spaces. In the optimal separable spaces, the differences in relative locations and densities between the target and the background are quantified by developing a specialized measurement for the detection output to satisfy the detection response requirements of highlighting the target while suppressing the background. The theoretical novelties and contributions of the research work in this article can be summarized as follows.

1) By adopting point-set topology theories to directly mine the differences between the target and the background in HSI data at the mathematical-statistical level, the limitations of the traditional modes of analyzing data through specific model assumptions are exceeded, thus gaining the flexibility to adapt to real scenes with variability to achieve high-precision detection.
2) For the rearrangement of data in topological spaces, information entropy estimation is innovatively introduced to measure the degree of ordering for the systems. Accordingly, spaces with the smallest uncertainty in the rearrangement of target and background are selected as the optimal separable spaces to ensure an accurate separation between them.
3) By efficiently extracting the land cover information for detection in the optimal separable spaces, a measurement of separation that can accurately quantify and highlight the differences in data features of various land covers is developed and utilized in the algorithmic output. With the guarantee of detection effect, while emphasizing computational efficiency, the proposed method is quite friendly to the quick response to the requirements of high-precision detection in real scenes.

The structure of this article is organized as follows. Section II describes in detail the methodology of the proposed method. Experimental results and the corresponding analysis and discussion are provided in Section III. Section IV summarizes the research work of this article and gives conclusions.

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II. Target Detection Through Information Entropy Estimation Based on Point-Set Topology

Fig. 1 provides the processing flow for the proposed TD-IEEPST, whose methodology can be described in three parts: 1) constructing parallel topological spaces by traversing all the bands of the HSI being processed to achieve an ordered arrangement of data in each space; 2) unraveling the data arrangement in different topological spaces by information entropy estimation so as to select the optimal separable spaces; and 3) in the optimal separable spaces, the detection output is developed based on the properties of the minimum shared point set between the testing pixel and the given target.

A. Construction of Parallel Topological Spaces

With the availability of the priori information about the target, supervised target detection is dedicated to finding pixels with high spectral similarity to the given target and realizing image-level diagnosis for the imaging scene, which can be treated as an information retrieval task in machine learning. Uneven distribution of land covers in the imaging scene is the basic condition for implementing detection. Theoretically, mathematical modeling of HSIs that satisfy the detection condition can yield a data cloud that contains datasets with different attributes. The coexistence of dense and sparse sets is a mathematical-statistical property that needs to be satisfied by the dataset on which the detection is implemented. An HSI, as a specific form of a collection of data points, point-set theories are perfect for analyzing its mathematical-statistical properties. The specificity of a target of interest is characterized by the fact that it is different from the surrounding background in terms of quantity, density, spectral characteristics, etc. Such specificity does exist, but it is a latent, hidden, and difficult-to-quantify feature in the original image. Hence, topology can be utilized to disassemble HSI data. Through a specific form of mapping, after mapping the HSI being processed from the primitive space to a topological space, the latent, hidden, and difficult-to-quantify features in the primitive space will be transformed into intuitive properties in the topological space. Such properties are particularly easy to quantify and can greatly facilitate the design of detection algorithms. In summary, in this article, the point-set topology theories are utilized to analyze and disassemble the data on the basis of which the requirements for detection response are met.

The construction of a topological space goes through the generation of point sets. The specific form of mapping from the primitive space to the topological space, or rather, the way of generating the point sets, should be conducive to highlighting the differences in data features between various classes of land covers. In a 3-D data cube corresponding to an HSI, the differences in data features of pixels belonging to various classes of land covers in terms of quantity, density, spectral trend, and morphology are essentially the numerical differences between various classes of spectra in specific bands. Thus, the generation of point sets in the topological space relies on numerical differences in the spectra corresponding to different pixels.

Assuming $\mathbf{X} \in \mathbb{R}^{L\times N}$ is an HSI being processed, L denotes the number of bands while N denotes the number of pixels. With N_{topo} denoting the number of point sets in a single topological space, the minimum-maximum interval of the HSI being processed is divided equally into N_{topo} parts, each of which corresponds to a central value. On a given band, all pixels are assigned to the corresponding intervals according to their respective closest center values and N_{topo} point sets are generated, whereby a topological space is constructed. For a given point set, the number of elements contained in it is defined as its cardinality. The sum of the cardinalities of N_{topo} point sets in a single topological space is equal to N. Traversing all the bands in the HSI, the same operation is performed to complete the construction of L parallel topological spaces. In the parallel topological spaces shown in Fig. 2, $N_{\text{topo}} = 16$.

On each band, with the construction of the topological space, N pixels are rearranged once. Pixels within the same point set are close to the same central value, and the set of N_{topo} central values is a set of equidistant series, whereby the HSI being processed undergoes an ordered rearrangement based on numerical differences in each topological space, which is an entropy-reduction process. The rearrangement of data on different bands varies, resulting in varying information entropies for the corresponding parallel topological spaces. As a basic concept in information theory, information entropy can measure the degree of ordering of the topological space as a system, giving the uncertainty of data rearrangement in it. Therefore, in the proposed method, the information entropy estimation is utilized to unravel the data arrangement in the topological space and extract the land cover information for detection. It is worth noting that the prior information about the target is not utilized in the construction of parallel topological spaces. Such an arrangement is beneficial for rapid construction to ensure computational efficiency. The subsequent selection of the optimal separable spaces makes full use of the prior information, which is advantageous for the accurate target-background separation to improve detection accuracy.

B. Selection of Optimal Separable Spaces With Information Entropy Estimation

In the parallel topological spaces, all pixels in the HSI being processed are rearranged around the same set of central values. Given that the same pixel exhibits different spectral characteristics, in other words, different numerical values, on different bands, the arrangement of pixels in the topological spaces corresponding to different bands varies. Taking a given target spectrum $\mathbf{t} \in \mathbb{R}^{L\times 1}$ as a priori information, it is possible to track which point set it lies within based on the value of $\mathbf{t}$ in each band. Given that target detection is essentially a binary classification problem, binary classification can be accomplished once for all pixels in each topological space based on the point set where $\mathbf{t}$ is located and the remaining $N_{\text{topo}} - 1$ point sets. On this basis, the introduction of information entropy estimation provides a powerful tool for unraveling the data arrangement in different topological spaces.

In a topological space, all pixels are dichotomized based on the point set where the target $\mathbf{t}$ is located, whereby random

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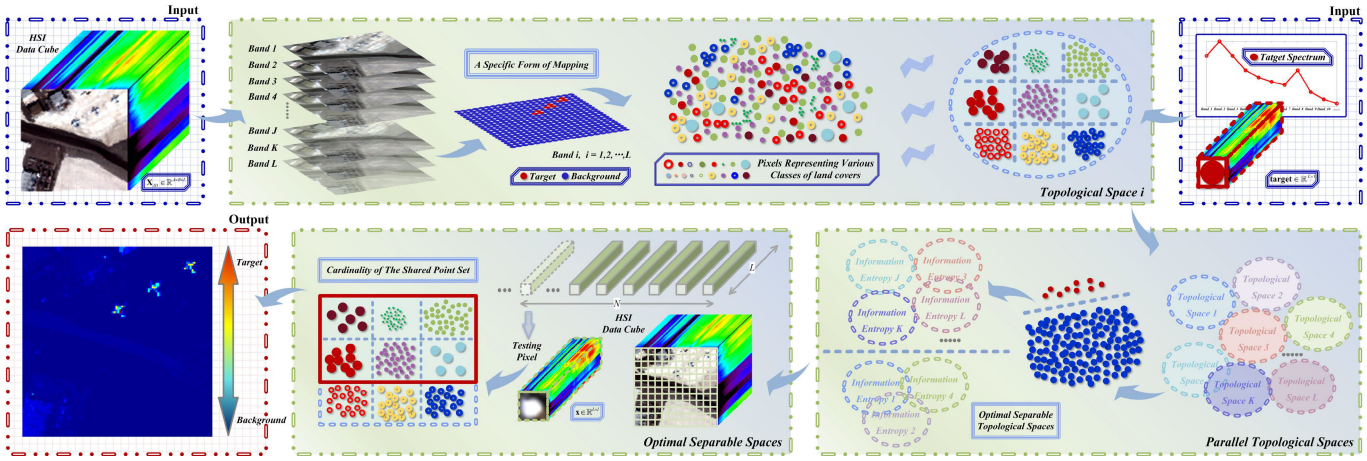


Fig. 1. Flowchart of the proposed TD-IEEPST for hyperspectral target detection.

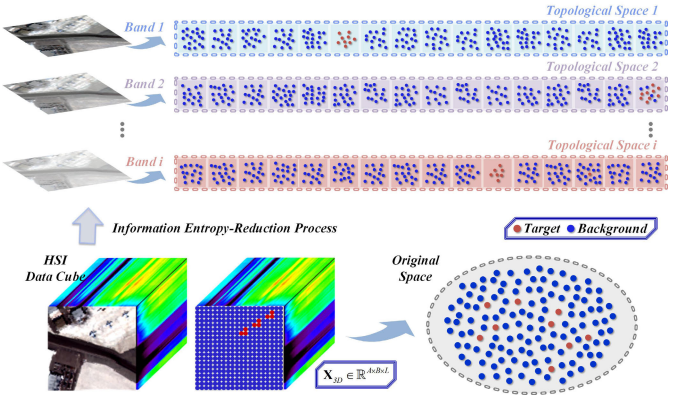


Fig. 2. Illustration of the construction of parallel topological spaces.

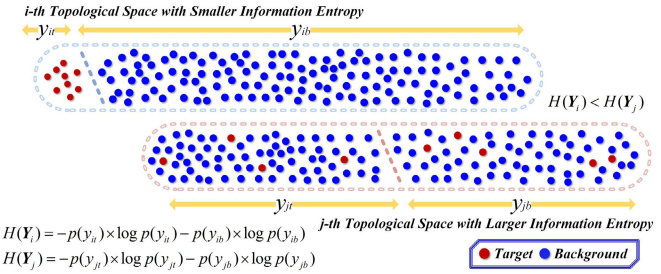


Fig. 3. Topological spaces with varying information entropies.

events are generated. And $\boldsymbol{Y}_i$, $i = 1, 2, \ldots, L$ represents a collection of information sources, in other words, a set of random events. The ordinal number of topological spaces is denoted by i and the total number is equal to the number of bands L. In $\boldsymbol{Y}_i$, symbols y_{it}, $i = 1, 2, \ldots, L$ and y_{ib}, $i = 1, 2, \ldots, L$ denote two random events, respectively: in the ith topological space, a pixel is assigned within a point set that is the same as, or different from, the target $\mathbf{t}$. Symbols $p(y_{it})$ and $p(y_{ib})$ denote the occurrence probabilities of these events. According to the law of large numbers in probability theory, the probability of an event can be replaced by the frequency of the event when the number of experiments is large enough. For the proposed method, the frequency of random events can be calculated from the cardinalities of the corresponding point sets, so we have

$$p(y_{it}) = \frac{\mathrm{Card}_{it}}{N} \tag{1}$$

$$p(y_{ib}) = \frac{\mathrm{Card}_{ib}}{N} \tag{2}$$

where Card_{it} denotes the cardinality of the point set where $\mathbf{t}$ is located in the ith topological space and Card_{ib} denotes the sum of the cardinalities of the remaining $N_{\mathrm{topo}} - 1$ point sets, $\mathrm{Card}_{it} + \mathrm{Card}_{ib} = N$. Accordingly, a probability space $\boldsymbol{P}_i, i = 1, 2, \ldots, L$ for $\boldsymbol{Y}_i$ can be obtained

$$\begin{bmatrix} \boldsymbol{Y}_i \\ \boldsymbol{P}_i \end{bmatrix} = \begin{bmatrix} y_{it} & y_{ib} \\ p(y_{it}) & p(y_{ib}) \end{bmatrix}. \tag{3}$$

According to the formula proposed by C.E. Shannon, the expression of information entropy is given as follows:

$$\begin{aligned} H(\boldsymbol{Y}_i) &= E(I(\boldsymbol{Y}_i)) = E\left[-\log p(\boldsymbol{Y}_i)\right] \\ &= -p(y_{it}) \times \log p(y_{it}) - p(y_{ib}) \times \log p(y_{ib}) \end{aligned} \tag{4}$$

where $I(\boldsymbol{Y}_i)$ denotes the information content of a random event that may occur in $\boldsymbol{Y}_i$, which is a measure of the uncertainty contained in the occurrence of the event. And the information entropy $H(\boldsymbol{Y}_i)$ denotes the average uncertainty contained in the occurrence of the events in $\boldsymbol{Y}_i$. The base of the logarithm is taken as 2 and the unit of information content is bit. As shown in Fig. 3, considering each topological space as a system, the larger the information entropy is, the less ordered and more unstable it is, in which the greater the uncertainty of data rearrangement, i.e., the greater the uncertainty of the division of all pixels in the HSI. Conversely, the more ordered and stable a system is, the less uncertainty there is in the data arrangement and the less uncertainty there is in the division of pixels.

As mentioned earlier, pixels belonging to various classes of land covers in the HSI being processed in topological spaces differ in their arrangement due to differences in deep-level data features. In a space with less information entropy, where the uncertainty of all pixels being dichotomized according

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to whether or not they are in the point set where the given target is located is lower, the difference in the arrangement of pixels belonging to various classes of land covers is more significant, which facilitates an accurate separation between the target and the background. Thus, among the parallel topological spaces, spaces with less information entropies are selected as the optimal separable spaces, where the specificity of the given target is more evident. Specifically, $\mathbf{H} = [H(\boldsymbol{Y}_1), H(\boldsymbol{Y}_2), \ldots, H(\boldsymbol{Y}_L)]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$ denotes the information entropies of L topological spaces, η_{ie} denotes the threshold used to select optimal separable spaces, and per_{ie} is a set percentage. With $H_{\max}$ and $H_{\min}$ denoting the maximum and minimum elements in $\mathbf{H}$, respectively, the threshold can be calculated as follows:

$$\eta_{ie} = H_{\min} + \mathrm{per}_{ie} \times (H_{\max} - H_{\min}). \tag{5}$$

The criteria for the selection of the optimal separable space are

$$\mathrm{OSS}_{\mathrm{TD}} = \{i \in \mathbb{N}^* | 0 < H(Y_i) < \eta_{ie}\} \tag{6}$$

where $\mathrm{OSS}_{\mathrm{TD}}$ denotes the set of ordinal numbers of the optimal separable spaces. The number of optimal separable spaces depends on the number of topological spaces with information entropies greater than 0 and less than η_{ie}. It is worth noting that the information entropy is nonnegative, when the information entropy is equal to 0, it means that in the corresponding topological space, there is an event in the random-event sets with an occurrence probability of 1 and another event with an occurrence probability of 0. In such a case, all pixels in the HSI being processed are divided into the same point set, which is not favorable for the separation between the target and the background, hence the case where the information entropy is equal to 0 is eliminated. The specific steps for the selection of the optimal separable spaces are given in Algorithm 1.

Algorithm 1 TD-IEE-OSS ($\mathbf{X}$, $\mathbf{t}$, N_{topo}, per_{ie})

Input: $\mathbf{X} = [\mathbf{X}_1; \ldots; \mathbf{X}_L] \in \mathbb{R}^{L\times N}$, where $\mathbf{X}_i \in \mathbb{R}^{1\times N}$-the HSI being processed, $\mathbf{t} = [t_1, \ldots, t_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$-the given target spectrum, N_{topo}-the number of point sets in each topological space, per_{ie}-the percentage used to set the threshold to constrain the information entropy.

Output: $\mathbf{Card} \in \mathbb{R}^{L\times N_{topo}}$-the cardinalities of point sets in parallel topological spaces, $\mathbf{Loca}_t \in \mathbb{R}^{L\times 1}$-the locations of point sets in parallel topological spaces where the given target is located, $\boldsymbol{OSS}_{TD}$-the set of ordinal numbers of the optimal separable spaces.

Initialization: $\mathbf{H} = [H_1, \ldots, H_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$, $\mathbf{Loca}_t = [Loca_{t1}, \ldots, Loca_{tL}]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$, $\mathbf{Card} = [\mathbf{Card}_1; \ldots; \mathbf{Card}_L] \in \mathbb{R}^{L\times N_{topo}}$, where $\mathbf{Card}_i = [Card_{i1}, \ldots, Card_{iN_{topo}}] \in \mathbb{R}^{1\times N_{topo}}$, $\boldsymbol{OSS}_{TD} = \emptyset$.

1: The minimum-maximum interval of $\mathbf{X}$ is divided equally into N_{topo} parts, each of which corresponds to a central value, and this set of central values is recorded in $\mathbf{Cen} \in \mathbb{R}^{1\times N_{topo}}$;

2: Start a traversal to construct parallel topological spaces and to estimate the information entropy for the data rearrangement in them;

for $i = 1 : L$ **do**

3: Denote the set of random events that is the data rearrangement in the current topological space by Y_i, $H(Y_i) = 0$;

4: Given $\mathbf{X}_i$ and $\mathbf{Cen}$, all pixels in the current band are assigned to the corresponding intervals according to their closest central values and N_{topo} point sets are generated;

5: The cardinalities of N_{topo} point sets are recorded in $\mathbf{Card}_i$;

6: Find the point set where t_i is located in the current topological space and record its location and cardinality in $Loca_{ti}$ and $Card_{it}$, respectively;

7: Denote by y_{it} and y_{ib} the two random events of whether a pixel in the current topological space is divided within the point set where $\mathbf{t}$ is located. Calculate the occurrence probabilities of y_{it} and y_{ib} and their information contents, $p(y_{it}) = \frac{Card_{it}}{N}$, $I(y_{it}) = -\log p(y_{it})$, $p(y_{ib}) = \frac{Card_{ib}}{N} = \frac{N - Card_{it}}{N} = 1 - p(y_{it})$, $I(y_{ib}) = -\log p(y_{ib})$;

8: $H_i = H(Y_i) = p(y_{it}) \times I(y_{it}) + p(y_{ib}) \times I(y_{ib})$;

9: **end for (step 2)**

10: Take the maximum and minimum elements of $\mathbf{H}$, $H_{\max}$ and $H_{\min}$, and set the threshold using per_{ie}, $\eta_{\mathrm{ie}} = H_{\min} + per_{ie} \times (H_{\max} - H_{\min})$;

11: Find ordinal numbers of the optimal separable spaces by means of the selection criteria, $OSS_{TD} = \{i | 0 < H(Y_i) < \eta_{\mathrm{ie}}\}$;

12: **return Card**, $\mathbf{Loca}_t$, OSS_{TD}

C. Shared Point Set in Optimal Separable Spaces for Detection Output

For the detection output, the similarity between a testing pixel and the given target is measured by intuitive properties in the optimal separable spaces. In each optimal separable space, all pixels are rearranged around the same set of equidistant series to achieve an ordered rearrangement based on numerical differences. Pixels belonging to various classes of land covers differ significantly in their arrangement, including relative locations and densities, resulting in them being located within clearly distinguishable point sets. Based on this, the similarity between a testing pixel and the given query is quantified by taking the given target spectrum as a given query and its manifestation in the optimal separable space as a reference, which provides a feasible strategy for the implementation of the information retrieval task.

Specifically, the construction of a topological space is essentially the generation of multiple point sets through a specific form of mapping in order to enhance the separability between data points with different mathematical-statistical properties. The mapping from the primitive space to the topological space achieves an ordered arrangement of data, which is an entropy-reduction process. In the optimal separable spaces, the point sets are arranged according to the rule of increasing central values, hence, the relative locations and densities between the point sets in which the testing pixel and the given target are

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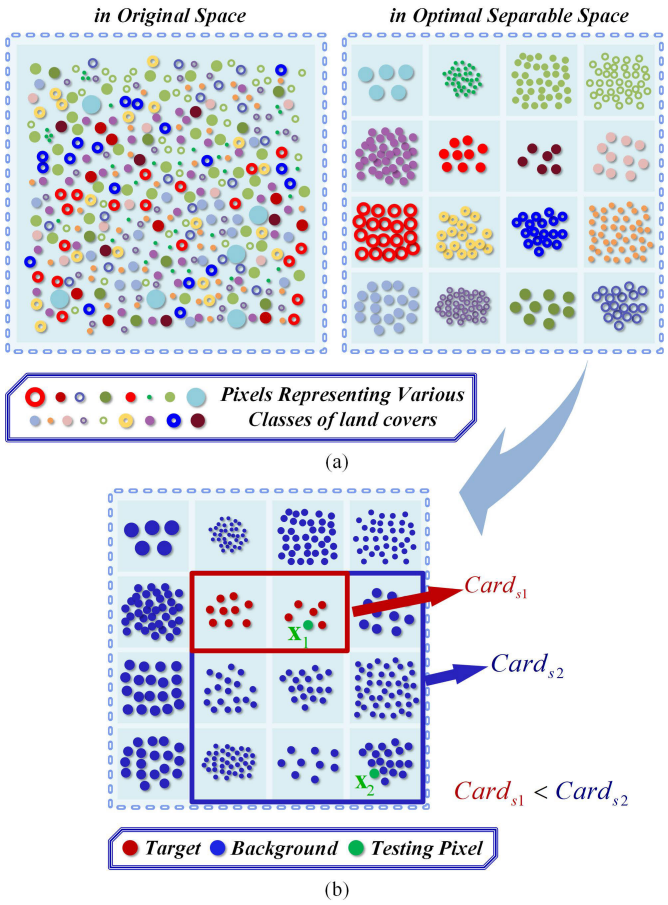


Fig. 4. Illustration of data arrangements. (a) Primitive space and the optimal separable space. (b) Shared point sets between testing pixels and the given target.

located can reflect the similarity between them. Specifically, the proposed method achieves the essential requirement of highlighting the target while suppressing the background by seeking point sets shared by the testing pixel and the given target in the optimal separable spaces, whose cardinalities are utilized in the detection output.

In a topological space, the shared point set between two pixels needs to fulfill the condition: with the inclusion of these two pixels, the number of point sets contained therein reaches a minimum. Notably, the shared point set between two pixels in a topological space is uniquely determined. For an HSI containing pixels belonging to various classes of land covers, Fig. 4(a) illustrates their arrangements in the primitive space and in the optimal separable space. Assuming that $\mathbf{x}_1$, $\mathbf{x}_2 \in \mathbb{R}^{L\times 1}$ are two testing pixels in $\mathbf{X}$, in the optimal separable space shown in Fig. 4(b), the shared point set between $\mathbf{x}_1$ and $\mathbf{t}$ is labeled with a red box and its cardinality is Card_{s1}. While the shared point set between $\mathbf{x}_2$ and $\mathbf{t}$ is labeled with a blue box with a cardinality of Card_{s2}, $\text{Card}_{s1} < \text{Card}_{s2}$. The smaller cardinality of the shared point set between $\mathbf{x}_1$ and $\mathbf{t}$ in topological space represents a higher similarity between the two in terms of their arrangements. Whereas, a larger cardinality of the shared point set between $\mathbf{x}_2$ and $\mathbf{t}$ in the topological space represents a lower similarity between the two. Thus, the cardinality of the shared point set between a data point and a given query can reflect the relevance of that point to the given query.

Denote a testing pixel in the HSI being processed by $\mathbf{x} \in \mathbb{R}^{L\times 1}$, shared point sets of $\mathbf{x}$ and $\mathbf{t}$ in the N_{oss} optimal separable spaces in OSS_{TD} are tracked and recorded in $\mathbf{Card}_{\text{share}} = [\text{Card}_{s1}, \ldots, \text{Card}_{sN_{\text{oss}}}]^{\text{T}} \in \mathbb{R}^{N_{\text{oss}}\times 1}$. In the jth optimal separable space, the locations of the point sets where $\mathbf{x}$ and $\mathbf{t}$ are located are recorded in Coor_{xj} and Coor_{tj}, respectively. With $\mathbf{Card}$, the cardinalities $\mathbf{Card}_j$ in the current optimal separable space are obtained. As mentioned earlier, all pixels are rearranged around the same set of equidistant series thus achieving an ordered rearrangement, and hence, the shared point sets can be localized by Coor_{xj} and Coor_{tj}. For Coor_{xj} and Coor_{tj}, assuming that Coor_{ja} denotes the smaller of the two and Coor_{jb} denotes the larger of the two, the cardinality of the shared point set can be obtained as follows:

$$\text{Card}_{sj} = \text{sum}(\mathbf{Card}_j(\text{Coor}_{ja}:\text{Coor}_{jb})). \tag{7}$$

Accordingly, the average of the cardinalities of the shared point sets in the optimal separable spaces can be calculated

$$\text{Card}_{\text{share-ave}} = \frac{\|\mathbf{Card}_{\text{share}}\|_1}{N_{\text{oss}}}. \tag{8}$$

To make the detection output satisfy the requirement of highlighting the target while suppressing the background, $\text{Card}_{\text{share-ave}}$ is normalized and the negative exponential function is employed

$$D_{\text{TD-IEEPST}}(\mathbf{x}) = \exp\left(-\frac{\text{Card}_{\text{share-ave}}}{N}\right). \tag{9}$$

The judgment criteria for target detection are given below

$$D_{\text{TD-IEEPST}}(\mathbf{x}) \begin{cases} > \eta \rightarrow \text{target} \\ < \eta \rightarrow \text{background} \end{cases} \tag{10}$$

where η is a given threshold. For the testing pixel $\mathbf{x}$, if $D_{\text{TD-IEEPST}}(\mathbf{x}) > \eta$, then $\mathbf{x}$ is more likely to be judged as belonging to the same class as $\mathbf{t}$, i.e., the target class. If $D_{\text{TD-IEEPST}}(\mathbf{x}) < \eta$, then $\mathbf{x}$ is more likely to be judged as belonging to the background class. Algorithm 2 specifically describes the implementation steps for the proposed TD-IEEPST.

The complete implementation process of the proposed method consists of Algorithms 1 and 2, which have two key parameters as inputs: the number of point sets N_{topo} in each topological space and the percentage per_{ie} used to set the threshold to constrain the information entropy. In the following, the corresponding experimental results are shown and qualitative and quantitative analyses are given for the impact of the settings of these two parameters on the performance including detection effects and computational efficiency.

III. Experimental Results and Analysis

For the discussion of key parameters and the comparison of detection performance, experiments are carried out to comprehensively evaluate the detection effects and computational efficiency of the proposed method. The qualitative and quantitative metrics adopted for the evaluation are well-recognized

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Algorithm 2 TD-IEEPST ($\mathbf{X}$, $\mathbf{t}$, $\mathbf{Card}$, $\mathbf{Loca}_t$, OSS_{TD})

Input: $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$, $\mathbf{X} \in \mathrm{R}^{L\times N}$-the HSI being processed, $\mathbf{t} = [t_1, \ldots, t_L]^{\mathrm{T}} \in \mathrm{R}^{L\times 1}$-the given target spectrum, $\mathbf{Card} \in \mathrm{R}^{L\times N_{topo}}$-the cardinalities of point sets in parallel topological spaces, $\mathbf{Loca}_t \in \mathrm{R}^{L\times 1}$-the locations of point sets in parallel topological spaces where the given target is located, $OSS_{TD} = [OSS_1, \ldots, OSS_{N_{oss}}]^T \in \mathrm{R}^{N_{oss}\times 1}$-the set of ordinal numbers of the optimal separable spaces.

Output: $\mathbf{d} = [d_1, \ldots, d_N] \in \mathrm{R}^{1\times N}$-detection results.

Initialization:

$\mathbf{Card}_{share} = [Card_{s1}, \ldots, Card_{sN_{oss}}]^{\mathrm{T}} \in \mathrm{R}^{N_{oss}\times 1}$.

1: Get the number of elements N_{oss} in OSS_{TD}, i.e., the number of optimal separable spaces;

2: Start a traversal to calculate the detection output;

for $i = 1 : N$ **do**

3: $\mathbf{x}_i \in \mathrm{R}^{L\times 1} \leftarrow i$-th testing pixel of $\mathbf{X}$;

4: Initialize the record of locations of the current testing pixel in the optimal separable spaces, $\mathbf{Coor}_x = [Coor_{x1}, \ldots, Coor_{xN_{oss}}]^{\mathrm{T}} \in \mathrm{R}^{N_{oss}\times 1}$;

5: Start a traversal to track the distribution of the current testing pixel $\mathbf{x}_i$ and the given target $\mathbf{t}$ in the optimal separable spaces;

for $j = 1 : N_{oss}$ **do**

6: Extract the ordinal number of the current optimal separable space OSS_j;

7: Get the values of the testing pixel and the given target in the current optimal separable space, $x_{ij} = \mathbf{x}_i(OSS_j)$, $t_j = \mathbf{t}(OSS_j)$;

8: Find the point set in which x_{ij} is located and record its location in $Coor_{xj}$;

9: Find the point set in which t_j is located by $\mathbf{Loca}_t$, $Coor_{tj} = \mathbf{Loca}_t(OSS_j)$;

10: Through $\mathbf{Card}$, obtain the cardinalities of all point sets in the current optimal separable space, $\mathbf{Card}_j = \mathbf{Card}(OSS_j, :)$;

11: $Coor_{ja} = \min([Coor_{xj}, Coor_{tj}])$, $Coor_{jb} = \max([Coor_{xj}, Coor_{tj}])$;

12: $Card_{sj} = \mathrm{sum}(\mathbf{Card}_j(Coor_{ja} : Coor_{jb}))$;

end

13: **end for (step 5)**

14: Calculate the average of the cardinalities of the shared point sets in the optimal separable spaces,

$$Card_{share-ave} = \frac{\|\mathbf{Card}_{share}\|_1}{N_{oss}};$$

15: With $Card_{share-ave}$ is normalized, the negative exponential function is employed for the detection output, $d_j = \exp(-\frac{Card_{share-ave}}{N})$;

end

16: **end for (step 2)**

17: **return d**

and widely used in the field to allow scientific and reasonable analysis and validation of the experimental results. Section III-A describes the experimental preparation including benchmark datasets, evaluation metrics, and platform configurations. Section III-B exhibits the experimental results for all the free parameters involved in the input of the proposed method and analyses them accordingly. In Section III-B1, the proposed method participates in fair comparisons with other sophisticated, advanced, and classic methods to fully demonstrate its effectiveness and efficiency.

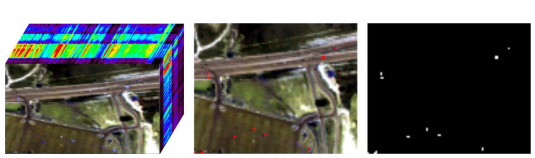
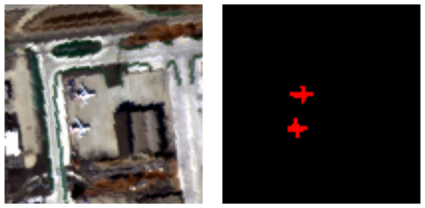
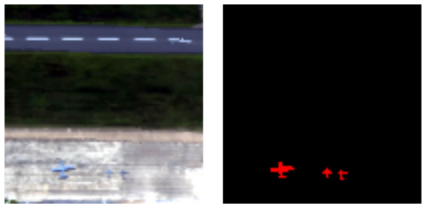
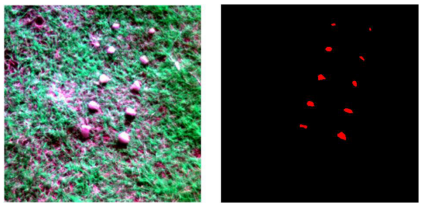
A. Experimental Preparation

1) Datasets: Six benchmark HSIs covering both natural and artificial environments are selected for experiments to sufficiently observe the performance of the proposed method. The details of the experimental dataset are given in Table I as follows.

1) The AVIRIS-1 was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) by imaging an urban area of San Diego at a spatial resolution of 3.5 m, with a spatial size of 100 × 100 [60]. The wavelength range of the spectral dimension is 370–2510 nm, and the number of bands is 224. A total of 189 bands were retained for the experiments after removing the absorption and low SNR bands. The targets of interest in the imaging scene are three airplanes, which occupy 22, 22, and 20 pixels, respectively.
2) The HYDICE was acquired through the hyperspectral digital imagery collection experiment (HYDICE) imaging at a spatial resolution of approximately 2 m for an area in Michigan, USA, with a spatial size of 80 × 100 [66]. There are 175 bands on the spectral dimension, and the targets of interest in the scene are vehicles distributed in ten different locations, each of which has a size of 1–4 pixels.
3) The Los Angeles was acquired by AVIRIS imaging of an area of Los Angeles, USA, with a spatial resolution of approximately 7.1 m [67]. The spatial size is 100 × 100. A total of 205 bands with good quality were retained for the experiments. The targets of interest are two airplanes, occupying 44 and 43 pixels, respectively.
4) The Gulfport was acquired by AVIRIS imaging of an area of Gulfport at a spatial resolution of 3.4 m [19]. The image contains 100 × 100 pixels and 191 bands. The targets to be detected in the scene are three airplanes occupying 10, 11, and 39 pixels, respectively.
5) The Cri was acquired by imaging an area of Wuhan University with the Nuance Cri hyperspectral sensor [68]. The full image contains 400 × 400 pixels with 46 spectral channels retained at a resolution of 10 nm. The targets of interest are various stones distributed in ten locations.
6) The WTC was acquired by NASA's Jet Propulsion Laboratory through AVIRIS imaging of the World Trade Centre (WTC) in New York [53]. The spatial size is 200 × 200. A total of 224 bands spanning 370–2510 nm in the spectral dimension were collected for the experiments. The targets of interest are fire sources occupying 83 pixels distributed in ten different locations.

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TABLE I
DETAILED INFORMATION OF EXPERIMENTAL DATASETS

	Spectro-meter	Capture Location	Spatial Resolution	Spatial Size	Number of Bands		Spectro-meter	Capture Location	Spatial Resolution	Spatial Size	Number of Bands
AVIRIS-1	AVIRIS	San Diego	3.5 m	100×100	189	HYDICE	HYDICE	Michigan	2m	80×100	175
											
Los Angeles	AVIRIS	Los Angeles	7.1m	100×100	205	Gulfport	AVIRIS	Gulfport	3.4m	100×100	191
											
Cri	Nuance Cri	Wuhan University	\	400×400	46	WTC	AVIRIS	New York	\	200×200	224
											

AVIRIS: Airborne Visible/Infrared Imaging Spectrometer HYDICE: Hyperspectral Digital Imagery Collection Experiment

2) Evaluation Metrics: The evaluation metrics adopted for the experimental results are well recognized and widely used in this field for scientific qualitative and quantitative analysis.

1) Given a set of thresholds, the receiver operating characteristic (ROC) curve is plotted with the false alarm rates (FARs) as the abscissa and the probabilities of detection (PDs) as the ordinate to show the detection effects of a method. The more superior the method, the more the ROC curve is skewed to the upper left.
2) The area under the curve (AUC) is a quantitative metric obtained by integrating the ROC curve. AUC provides a more intuitive, quicker, and accurate way of distinguishing subtle differences in performance between various methods. The stronger the detection capability of a method, the larger its AUC.
3) Based on the location information provided by the ground truth about the target and the background in a given scene, the detection results can be divided into two sets of values. The box-and-whisker separability map provides a visualization of the ability of a method to separate the target and the background by comparing the statistical ranges of these two sets of values. The lower the intersection degree between two colored boxes, the better the separation ability of the corresponding method.

3) Experimental Configurations: To be fair, all experiments in this article were conducted on the same software-hardware configuration, which is MATLAB R2014a on a Windows 11 computer with a 12th Gen Intel[1] Core[2] i7-1260 CPU @ 2.10 GHz and 16.0 GB RAM.

B. Discussion on Parameters

For the proposed TD-IEEPST, the algorithmic input contains two key parameters. N_{topo} determines the number of point sets in each topological space, while per_{ie} determines the threshold to constrain the information entropy. To observe the performance of the proposed method as the parameters are varied and to give reasonable recommendations, the settings of N_{topo} and per_{ie} are both selected to be in certain ranges. Specifically, the range for N_{topo} is [69 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200], while the range for per_{ie} is [10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%]. It is worth noting that there is no specific constraint between the setting of N_{topo} and the number of pixels in an HSI being processed. In the construction of a single topological space, all pixels in a specific band are assigned to the corresponding interval according to their closest central value and N_{topo} point sets are generated. Therefore, the setting of N_{topo} is not limited by the number of pixels in the HSI. The

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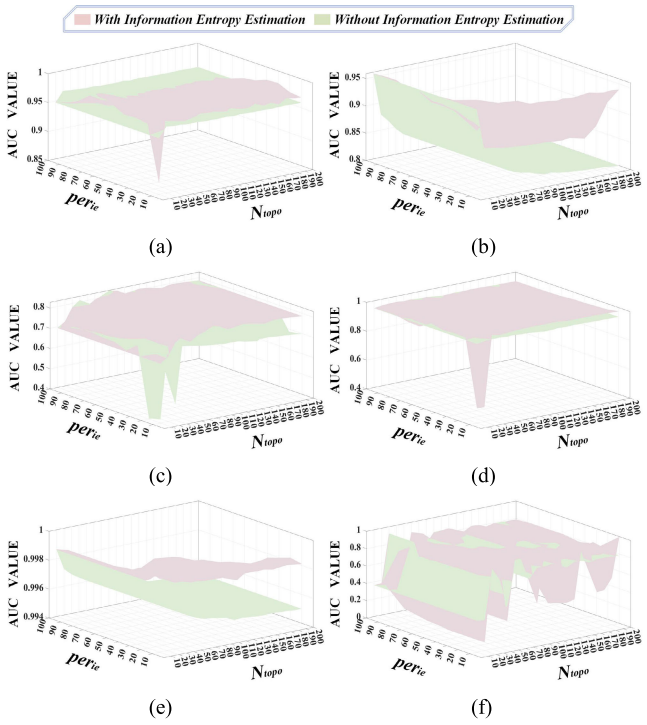


Fig. 5. Monochromatic surfaces illustrating the relationship between different parameter combinations and the detection performance for two sets of detection outputs, whether the optimal separable spaces are selected with information entropy estimation or not, on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri. (f) WTC.

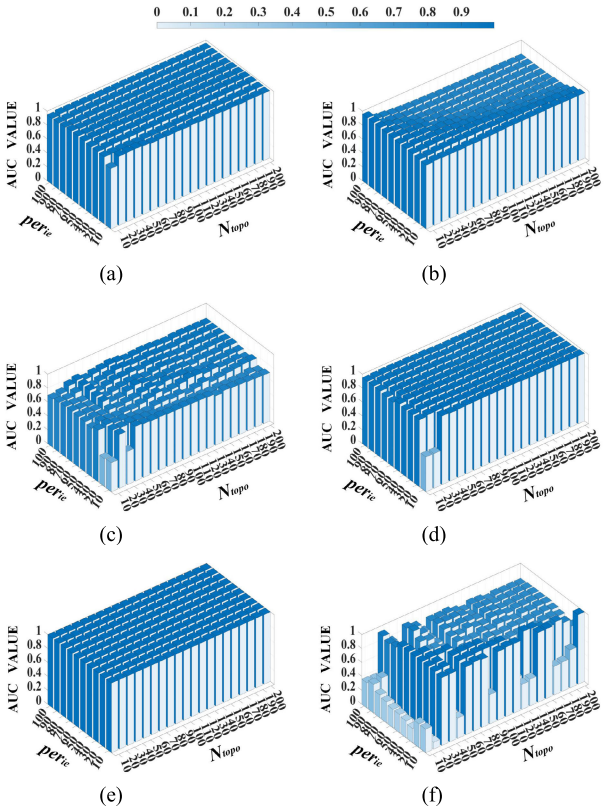


Fig. 6. Colored bar charts illustrating the relationship between combinations of N_{topo} and per_{ie} and the detection effect of TD-IEEPST on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri. (f) WTC.

metric used to evaluate the detection results for the parameter discussion is AUC. Extensive experiments are performed on each experimental dataset with the same parameter setting ranges and the execution time is recorded to avoid errors in experimental processes affecting the conclusions. This allows a comprehensive evaluation of the proposed method in terms of detection effects and computational efficiency.

1) Ablation Analysis: In order to fully verify that the proposed method using information entropy estimation to select the optimal separable spaces can achieve accurate target-background separation and improve the detection accuracy, the ablation analysis is carried out under the premise of constructing parallel topological spaces: the detection outputs are evaluated and compared based on whether the optimal separable spaces are selected with information entropy estimation or not. Specifically, for each experimental dataset, for a total of 200 parameter combinations as described above, the detection outputs based on all parallel topological spaces and those based on the optimal separable space are evaluated using the quantitative metric of AUC and plotted on the same coordinate system as two monochromatic surfaces. Accordingly, it is intuitive to compare two detection effects corresponding to whether or not the optimal separable spaces are selected using information entropy estimation.

Fig. 5(a)–(f) gives the ablation experimental results on AVIRIS-1, HYDICE, Los Angeles, Gulfport, Cri, and WTC, respectively. It can be observed that on all datasets, despite local intersections, the AUC surfaces corresponding to the optimal separable spaces adopted by the proposed method demonstrate superior performance generally over the AUC surfaces corresponding to all parallel topological spaces. This indicates that the selection of optimal separable spaces with information entropy estimation is conducive to the accurate separation between the target and the background, which effectively improves the detection effects. In conclusion, the introduction of information entropy estimation enables the proposed method to effectively extract land cover information for detection.

TABLE II
RECOMMENDED PARAMETER SETTINGS FOR IEEPST ON EXPERIMENTAL DATASETS

	Evaluation Result (AUC)	Recommended Parameter Setting	
		N_{topo}	per_{ie}
AVIRIS-1	0.9967	40	10%
HYDICE	0.9590	10	80%
Los Angeles	0.8282	80	100%
Gulfport	0.9943	20	50%
Cri	0.9993	10	20%
WTC	0.9986	40	50%

2) Detection Effect: Fig. 6(a)–(f) illustrates the results of the parameter discussion on AVIRIS-1, HYDICE, Los Angeles, Gulfport, Cri, and WTC, respectively. With the coloring of the bar charts according to the magnitude of AUCs, the detection effects of the proposed method can be observed

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TABLE III
EVALUATION RESULTS FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

	MF	GLRT	CEM	ACE	S-ACE	CSCR	SLRMD	DM-BDL	TD-TSE	TD-IEEPST
AVIRIS-1	0.9340	0.9379	0.9298	0.9379	0.9347	0.9949	0.9838	0.9374	0.9950	**0.9967**
HYDICE	0.9649	0.8965	**0.9692**	0.8963	0.9476	0.9858	0.9671	0.9469	0.9685	0.9590
Los Angeles	0.7580	0.7268	0.7528	0.7265	0.7526	**0.9378**	0.8547	0.8720	0.9316	0.8282
Gulfport	0.9747	0.9520	0.9750	0.9520	0.9741	0.9929	0.9676	0.9226	0.9938	**0.9943**
Cri	0.9820	0.9246	0.9711	0.9234	0.9617	0.9962	**0.9997**	0.9925	0.9971	0.9993
WTC	0.8370	0.8630	0.8372	0.8627	0.8236	0.7958	0.9117	0.8729	0.8713	**0.9986**
Average	0.9084	0.8835	0.9059	0.8831	0.8991	0.9506	0.9474	0.9241	0.9596	**0.9627**

as the parameters are varied on the experimental datasets. Correspondingly, when the AUC is maximized on each dataset, the recommended parameter settings are given in Table II for reference. The following conclusions can be drawn as follows.

1) On AVIRIS-1, HYDICE, Gulfport, and Cri, the variation of AUCs is slight. This suggests that the proposed method is quite stable and exhibits robustness in detection within the given parameter setting ranges.
2) On Los Angeles and WTC, the magnitude of variation in AUCs is relatively large when N_{topo} and per_{ie} are small. With the increase of the two parameters, the variation of AUCs decreases gradually and finally converges. It is indicated that the proposed method exhibits low sensitivity to the parameters within reasonable setting ranges.
3) The integration of the experimental results reveals that the proposed method performs stably on HSIs with a relatively homogeneous distribution of imaging scenes such as AVIRIS-1, HYDICE, Gulfport, and Cri. Comparatively, on HSIs with relatively complex imaging scenes such as Los Angeles and WTC, the proposed method also shows robustness after the parameter tuning process. This is a strong demonstration that the proposed method is quite user-friendly for locating targets of interest and meeting detection response requirements in imaging scenes with varying degrees of complexity.

3) Computational Efficiency: In addition to the evaluation of the detection results using AUC, the execution time of the proposed method is recorded on all experimental datasets. In Fig. 7(a)–(f), the surfaces are colored using the magnitude of time to demonstrate the computational efficiency for TD-IEEPST. It can be observed that the surfaces on all experimental datasets are close to sloping, indicating that the time complexity of the proposed method is close to linear. The low computational cost of TD-IEEPST makes it applicable to large-scale HSIs in practical applications. Moreover, among the two key parameters, per_{ie} has a more significant effect on the computational efficiency than N_{topo}, which indicates that time resources can be saved to a certain extent during the parameter tuning process. In summary, the nearly linear time complexity facilitates the proposed method to quickly adapt to varying scenes and achieve high-precision detection.

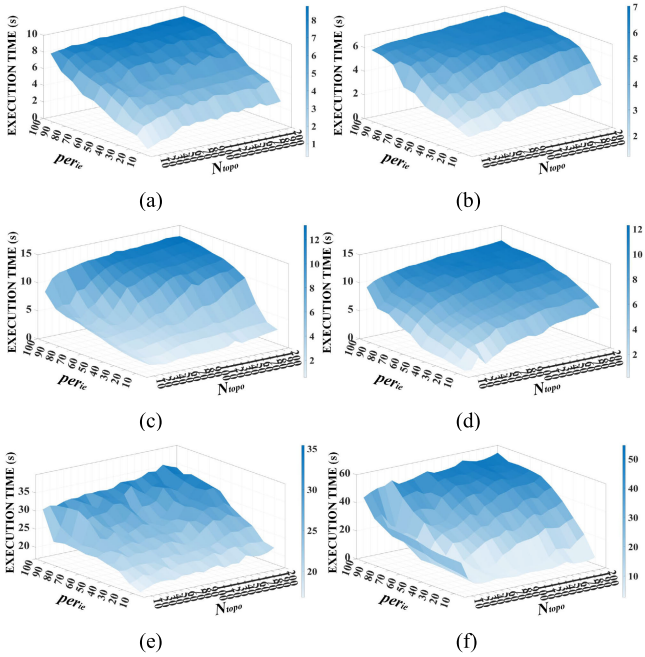


Fig. 7. Colored surfaces illustrate the relationship between combinations of N_{topo} and per_{ie} and the computational efficiency of TD-IEEPST on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri. (f) WTC.

C. Comparison of Performance

Nine classic and state-of-the-art hyperspectral target detection methods are used to compare with TD-IEEPST to further demonstrate the superior performance of the proposed method: MF [20]; GLRT [21]; CEM [70]; ACE [22]; signed ACE (S-ACE) [71]; CSCR [33]; SLRMD [34]; decomposition model with background dictionary learning (DM-BDL) [72]; and target detection through tree-structured encoding (TD-TSE) [42]. It can be seen that part of the methods in the comparison adopt the concentric dual-window detection mode, hence the size of the window is set appropriately according to the target information in the experimental datasets described in Section III-A to ensure that the detection methods achieve satisfactory effects. The inner and outer windows of AVIRIS-1 are set to 5×5 and 15×15, respectively. And the windows of HYDICE are set to 3×3 and 13×13. For Los Angeles and Gulfport, the inner and outer windows are 7×7 and 17×17. The window sizes of Cri are 15×15 and 25×25. The window sizes of WTC are 5×5 and 15×15.

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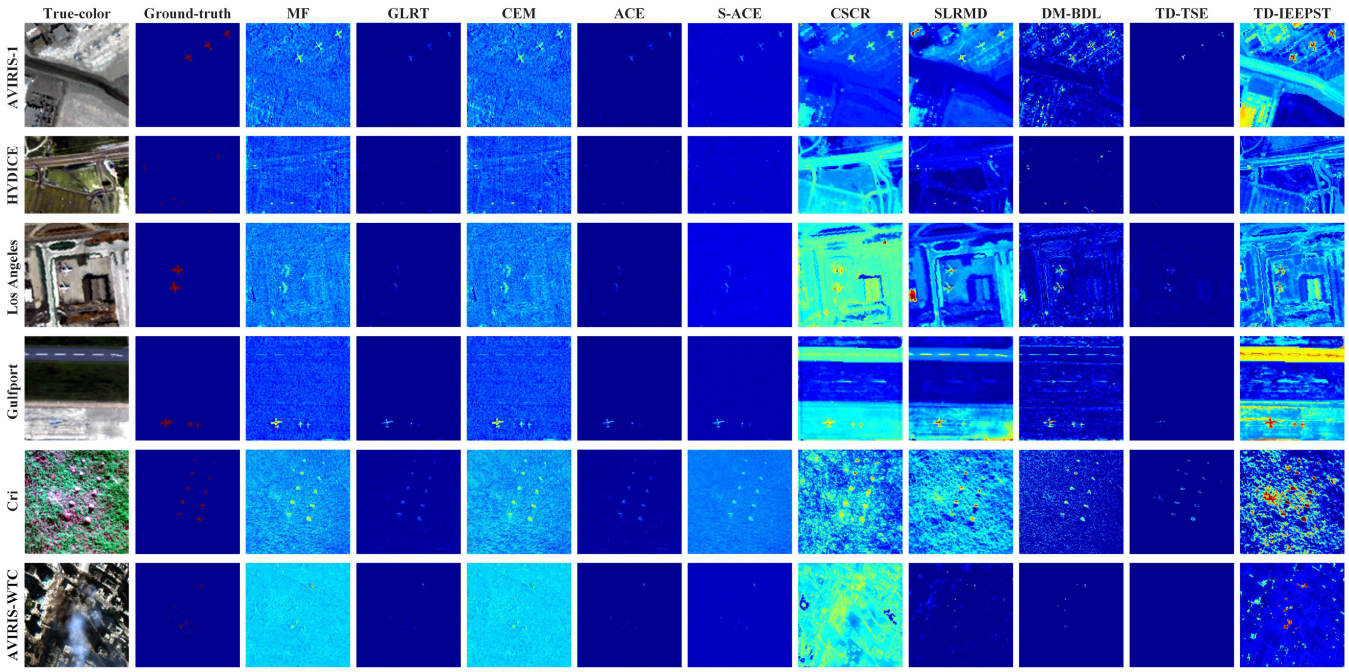


Fig. 8. Colored maps for comparison between detection methods on experimental datasets.

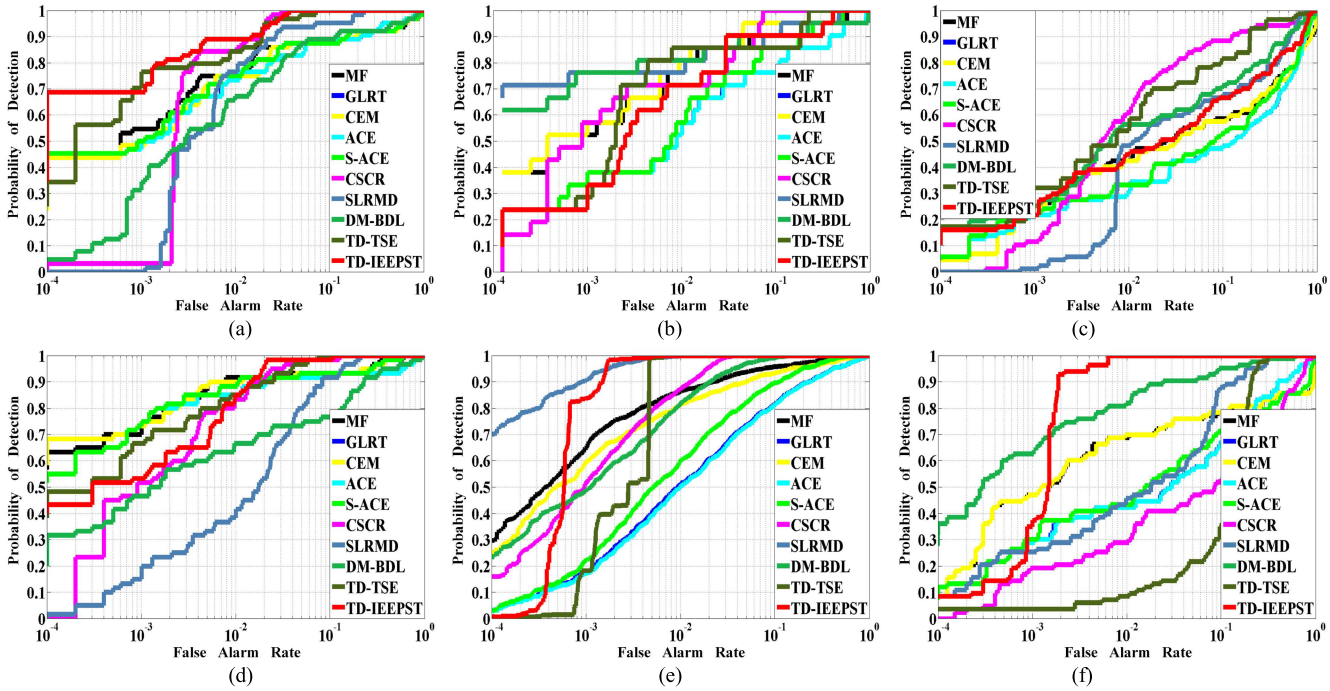


Fig. 9. ROC curves for comparison between detection methods on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri. (f) WTC.

1) Detection Effect: For the comparison, the detection results of all methods are linearly stretched to 0–255 and colored according to the magnitude of values, as shown in Fig. 8. Conclusions can be drawn by observing the visual results of the detection methods.

1) In terms of highlighting targets, GLRT, ACE, S-ACE, and TD-TSE performed poorly, with poorly delineated shapes and contours of targets in the detection results, indicating overall low values at the target locations.
2) In contrast, MF, CEM, CSCR, SLRMD, and DM-BDL are more capable of highlighting targets, and the targets are more clearly presented in the detection maps. However, pixels at the target locations of these five methods cannot be clearly distinguished in the results on WTC either.
3) Overall, TD-IEEPST exhibits the strongest capability for highlighting targets. Referring to the ground truth of the experimental datasets, it can be seen that values at the target locations in the detection results of the proposed

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TABLE IV
WILCOXON RANK SUM TEST FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

	MF	GLRT	CEM	ACE	S-ACE	CSCR	SLRMD	DM-BDL	TD-TSE	TD-IEEPST
AVIRIS-1	1	4	0	4	2	7	6	3	8	9
HYDICE	5	1	8	0	3	9	6	2	7	4
Los Angeles	4	1	3	0	2	9	6	7	8	5
Gulfport	5	1	6	1	4	7	3	0	8	9
Cri	4	1	3	0	2	6	9	5	7	8
WTC	2	5	3	4	1	0	8	7	6	9
Sum	21	13	23	9	14	38	38	24	**44**	**44**

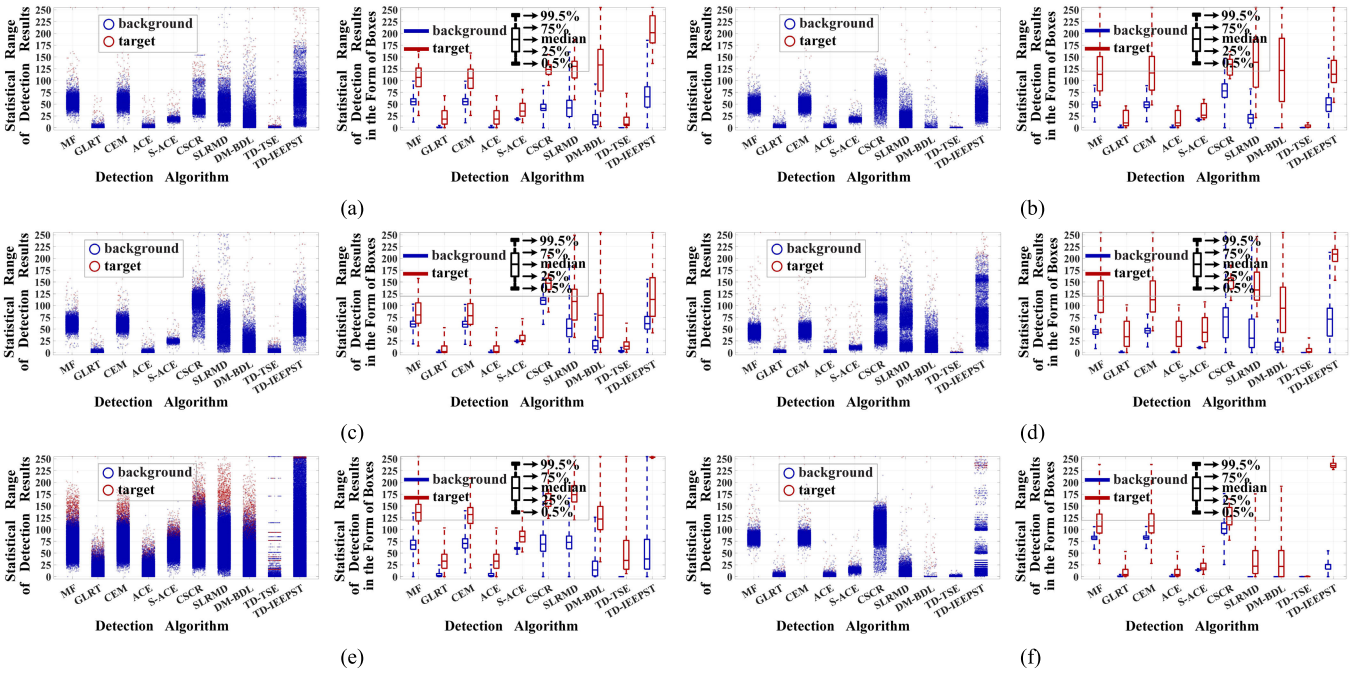


Fig. 10. Separability map for comparison between detection methods on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri. (f) WTC.

TABLE V
TIME COMPLEXITY FOR COMPARISON BETWEEN DETECTION METHODS

Method	Computational Complexity	Definition of Variables
MF	$O(L^3+L^2N)$	L : the number of bands N : the number of pixels *winout* : the size of the outer window *maxIter* : the number of iterations m : the number of background atoms $tree_{num}$: the number of trees $tree_{size}$: the subsampling size
GLRT	$O(L^3+L^2N)$	
CEM	$O(L^3+L^2N)$	
ACE	$O(L^3+L^2N)$	
S-ACE	$O(L^3+L^2N)$	
CSCR	$O(L^3Nwinout^2+LN+N)$	
SLRMD	$O(maxIter(L^2N+LN+N))$	
DM-BDL	$O(maxIterN+mN(L+1))$	
TD-TSE	$O(tree_{num}(tree_{size}+2N))$	
TD-IEEPST	$O(LN+LN_{topo}+NN_{oss})$	

method are at a relatively high level. With the shapes and contours of the targets of interest preserved more completely, the proposed method presents outstanding visual effects.

Fig. 9(a)–(f) plots ROC curves for all methods for further analysis of detection effects. Given the existence of ROC curves that intersect with each other in a way that is difficult to distinguish, Table III provides AUC values for the ROC curves in Fig. 9. On each experimental dataset, the AUC value corresponding to the best-performing method is marked in bold. For each method, the last row of the table shows the average of the AUCs over all datasets. In addition, given the large number of experimental results used for comparison, the Wilcoxon rank sum test is employed to provide an integrated and comprehensive evaluation [73]. The rows and columns of Table IV are labeled with the ten detection methods and the six experimental datasets, respectively. Referring to Table III, compare the AUC of each corresponding location with the remaining nine AUCs of the horizontal row in which it is located, and if it is greater, one point is awarded, otherwise no points are given. By analogy, the assignment of scores for all spaces in Table IV is completed. The last row of the table is the sum of the scores of each method on all the datasets, i.e., the Wilcoxon scores of all the methods. A larger Wilcoxon score indicates a better overall performance

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TABLE VI
EXECUTION TIME FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

	MF	GLRT	CEM	ACE	S-ACE	CSCR	SLRMD	DM-BDL	TD-TSE	TD-IEEPST
AVIRIS-1	0.1165	0.2871	0.1502	0.2337	0.3608	15.7348	42.3967	7.4505	7.7228	1.3626
HYDICE	0.0984	0.2643	0.0863	0.1611	0.2349	8.6802	30.1088	7.8784	5.0624	5.3125
Los Angeles	0.1534	0.3163	0.1091	0.3467	0.3637	23.9587	58.5293	9.1781	5.4508	8.9805
Gulfport	0.1250	0.3632	0.1502	0.2874	0.4063	28.3744	50.6141	8.1495	6.0325	2.7334
Cri	0.4784	1.0094	0.4494	0.8118	1.0394	743.5441	254.9407	95.7412	223.0400	18.8630
WTC	0.5047	1.0186	0.4044	0.7957	0.9689	87.8089	308.4575	20.5563	175.9355	3.6605
Sum	1.4764	3.2589	1.3496	2.6364	3.3740	908.1011	745.0471	148.9540	423.2440	40.9125

of the corresponding method on all experimental datasets. Summarizing the experimental results leads to the following conclusions.

1) As shown in Fig. 9, the performance of TD-IEEPST consistently leads to AVIRIS-1 with a significant advantage. On Gulfport, Cri, and WTC, the PD of TD-IEEPST is the highest when FAR $>$ 0.02, leading the other nine methods overall.
2) Combined with the separability map with box-and-whisker in Fig. 10, by observing (b), (d), and (f) of Fig. 9, it can be obtained that when FAR $<$ 0.01, the overall performance of TD-IEEPST is slightly inferior due to the intersection of the statistical ranges of the values at target and background locations in the detection results. Also, the poor performance of the ROC curve of TD-IEEPST in Fig. 9(c) also echoes the evaluation results in Fig. 10(c).
3) In Table III, TD-IEEPST has the most superior AUC evaluations on AVIRIS-1, Gulfport, and WTC. On Cri, TD-IEEPST is second only to SLRMD. And on HYDICE and Los Angeles, the performance of TD-IEEPST is relatively impressive even though its AUCs are not the maximum. Moreover, TD-IEEPST exhibits the best overall performance with the largest average of AUCs across all experimental datasets.
4) In Table IV, the Wilcoxon scores of TD-TSE and TD-IEEPST are the largest among all the methods involved in the comparison, and TD-IEEPST has the largest average AUC. This further indicates that the proposed method achieves the best overall detection effects on all datasets. And the Wilcoxon scores for CSCR and SLRMD are equal, second only to TD-TSE and TD-IEEPST, which also demonstrate excellent detection capabilities.

Fig. 10(a)–(f) gives separability maps for all methods to present the separability between the target and the background for all detection results. In each subfigure, statistical ranges of the two sets of values are plotted directly on the left, and on the right in the form of colored boxes. Conclusions can be drawn from the observations as follows.

1) For GLRT, ACE, S-ACE, and TD-TSE, the overall level of the values at the target locations in the detection results is relatively low, resulting in an incapability to achieve effective highlighting for the targets of interest, which further explains the visual effects of the detection maps in Fig. 8.
2) For MF, CEM, CSCR, SLRMD, and DM-BDL, the intervals of values at the target locations are higher, indicating that these five methods are more capable of highlighting targets. Moreover, DM-BDL can suppress the detection values at the background locations to very low levels, which corresponds to its excellent visual effects in Fig. 8.
3) On AVIRIS-1, Gulfport, Cri, and WTC, the statistical range of values at the target locations for TD-IEEPST is relatively concentrated and the overall level is the highest among all methods. This indicates that the proposed method is ahead of other methods in its capability of highlighting targets. On HYDICE and Los Angeles, TD-IEEPST is also able to effectively separate the target from the background.

It is worth noting that the ten detection methods involved in the comparison can be broadly categorized into two groups according to the origin of the development field: the traditional signal processing-based MF, GLRT, CEM, ACE, S-ACE, and the machine learning-based CSCR, SLRMD, DM-BDL, TD-TSE, and TD-IEEPST. According to Table IV, it can be seen that the Wilcoxon scores of several machine learning-based methods are higher than those of traditional methods. This is a reflection of the fact that machine learning-based methods are overall ahead of traditional methods in terms of detection effects. Yet, the advantages of traditional methods such as low computational cost and fast response speed cannot be ignored. In practical applications, various categories of methods should be appropriately selected according to the specific requirements to complete the conversion from theory to practice, and to achieve the deepening and popularization of the research related to hyperspectral target detection.

2) Computational Efficiency: With the detection effects fully analyzed, and to evaluate the comprehensive performance of the methods participating in the comparison, Table V lists their approximate time complexities. Table VI records the execution time in seconds on each dataset, and the last row of the table sums the execution times on all datasets for comparison. It is worth noting that the time complexity in Table V is derived from the analysis of each method relative to its respective key parameters, emphasizing the relationship between execution time and input parameters.

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The ultimate execution time is related to the specific one-step operations in each method. It is undeniable that the processing speed of the traditional methods MF, GLRT, CEM, ACE, and S-ACE is remarkably fast. Combined with the previous analysis, it can be seen that TD-IEEPST achieves the highest computational efficiency among the methods belonging to the machine learning category that are involved in the comparison, with a total execution time much lower than that of CSCR, SLRMD, DM-BDL, and TD-TSE. In summary, the proposed method can be implemented without costly computation, which effectively saves space and time resources, on the premise that the detection effects are overwhelmingly ahead of the traditional signal processing-based methods. The superior comprehensive performance provides strong support for the proposed method to achieve high-precision and high-efficiency localization for targets of interest in real imaging scenes.

IV. Conclusion

Aiming at the incapability of traditional signaling models to deeply mine feature differences of land covers to achieve high-precision target detection, this article analyzes the data features of the target and the background in HSIs from the mathematical-statistical level. An TD-IEEPST is proposed. By constructing parallel topological spaces, the differences in deep-level data features between various land covers are intuitively presented, which facilitates information extraction for detection. Then the optimal separable spaces for the target and the background are selected by fully exploiting the priori spectrum and combining it with the information entropy estimation. On this basis, the algorithmic output is designed to accomplish the detection based on properties of shared point sets between the testing pixel and the given target in the optimal separable spaces. TD-IEEPST can be implemented with low computational cost while guaranteeing high-precision detection and exhibiting prominent comprehensive performance in imaging scenes with varying degrees of complexity. The proposed method interprets hyperspectral target detection from a brand-new perspective, which not only stimulates more possibilities for theoretical migration in related research, but also provides novel and robust solutions for various detection response requirements in practical applications.

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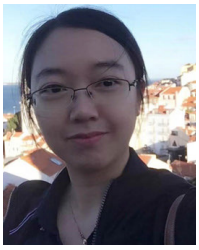
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Xiaotong Sun received the B.E. degree in electronic and information engineering from Beijing University of Chemical Technology, Beijing, China, in 2018, and the M.S.E. degree in information and communication engineering from the Airborne Remote Sensing Center, Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, in 2021. She is currently pursuing the Ph.D. degree in information and communication engineering with Hohai University, Nanjing, China.

Her research interests include signal processing, machine learning, hyperspectral image processing, and target detection.



Lina Zhuang (Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the Instituto Superior Tecnico, Universidade de Lisboa, Lisbon, Portugal, in 2018.

From 2015 to 2018, she was a Marie Curie Early Stage Researcher of Sparse Representations and Compressed Sensing Training Network (SpaRTaN number 607290) with the Instituto de Telecomunicacoes, Lisbon. SpaRTaN Initial Training Networks (ITNs) is funded under the European Union's Seventh Framework Programme (FP7-PEOPLE-2013-ITN) call and is part of the Marie Curie Actions-ITN funding scheme. From 2019 to 2021, she was a Research Assistant Professor with the Department of Mathematics, Hong Kong Baptist University, Hong Kong. From 2021 to 2022, she was a Research Assistant Professor with the Department of Mathematics, University of Hong Kong, Hong Kong. She is currently an Associate Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. Her research interests include hyperspectral image restoration, super resolution, and compressive sensing.



Lianru Gao (Senior Member, IEEE) received the B.S. degree in civil engineering from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 2007.

He was a Visiting Scholar at the University of Extremadura, Cáceres, Spain, in 2014, and Mississippi State University (MSU), Starkville, MS, USA, in 2016. He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS. In last ten years, he was the PI of ten scientific research projects at national and ministerial levels, including projects by the National Natural Science Foundation of China from 2018 to 2020, from 2022 to 2025, and from 2024 to 2028, and the National Key Research and Development Program of China from 2021 to 2025. He has published more than 240 peer-reviewed articles, and there are more than 150 journal articles included by Science Citation Index (SCI). He was a coauthor of three academic books including *Hyperspectral Image Information Extraction*. He obtained 30 National Invention Patents in China. He was supported by the National Science Foundation for Distinguished Young Scholars of China in 2023. His research focuses on hyperspectral image processing and information extraction.

Dr. Gao is a fellow of the Institution of Engineering and Technology. He received the Outstanding Science and Technology Achievement Prize of the CAS in 2016, the Second Prize of the State Scientific and Technological Progress Award in 2018, and the 2021 Outstanding Paper Award from the IEEE Workshop on Hyperspectral Image Processing: Evolution in Remote Sensing (WHISPERS). He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IET *Image Processing*.



Hongmin Gao (Member, IEEE) received the Ph.D. degree in computer application technology from Hohai University, Nanjing, China, in 2014.

He is currently a Professor with the College of Information Science and Engineering, Hohai University. His research interests include deep learning, information fusion, and image processing in remote sensing.



Xu Sun (Member, IEEE) received the B.S. degree in mathematics and application mathematics from Tsinghua University, Beijing, China, in 2006, and the Ph.D. degree in cartography and geographical information system from the Graduate University of Chinese Academy of Sciences, Beijing, in 2011.

He is an Associate Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include hyperspectral image processing, artificial intelligence algorithm, and high-resolution remote sensing image information mining.



Bing Zhang (Fellow, IEEE) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS, where he has been leading key scientific projects in the area of hyperspectral remote sensing for more than 25 years. He has developed five software systems in the image processing and applications. He has authored more than 300 publications, including more than 200 journal articles. He has edited six books/contributed book chapters on hyperspectral image processing and subsequent applications. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang has been serving as a Technical Committee Member for IEEE Workshop on Hyperspectral Image and Signal Processing since 2011. He is the Student Paper Competition Committee Member of IGARSS from 2015 to 2019. His creative achievements were rewarded ten important prizes from Chinese Government, and special government allowances of the Chinese State Council. He received the National Science Foundation for Distinguished Young Scholars of China in 2013, the 2016 Outstanding Science and Technology Achievement Prize of the Chinese Academy of Sciences, and the highest level of Awards for the CAS Scholars. He is currently serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. He has been serving as the President for the Hyperspectral Remote Sensing Committee of China National Committee of International Society for Digital Earth since 2012.

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